Watermarkable Speculative Permute-and-Flip Decoding

Setup. Let P and Q denote the target and draft language models, respectively, and let $u_c(y)$ and $v_c(y)$ be their processed logits for token $y \in \mathcal{V}$ at context c. The temperature is $T > 0$. A keyed generator $G(\mathsf{k}, c)$ returns a vector $E_c = (E_c(y))_{y \in \mathcal{V}}$ defined by

$$r_c(y) = \mathrm{PRF}_{\mathsf{k}}(c, y) \in (0, 1), \qquad E_c(y) = -\log r_c(y). \tag{1}$$

Over a uniformly random secret key, the values $r_c(y)$ are computationally indistinguishable from independent $\mathrm{Unif}(0, 1)$ variables, and hence $E_c(y)$ are computationally indistinguishable from independent $\mathrm{Exp}(1)$ variables. Thus, the secret key injects pseudorandom exponential noise and simultaneously provides the watermark.

1 Vanilla PF in report-exponential-noisy-max form

Algorithm 1: Watermarked Permute-and-Flip decoding via report exponential noisy max

Given: output length N, initial prompt $w_{1:n}$, target model P, temperature T, keyed exponential-noise generator G, and secret key k

while $n < N$ **do**
- $c \leftarrow w_{1:n}$; compute target logits $u_c(\cdot) \leftarrow P(c)$
- $E_c \leftarrow G(\mathsf{k}, c)$, where $E_c(y) = -\log \mathrm{PRF}_{\mathsf{k}}(c, y)$ for every $y \in \mathcal{V}$ // keyed pseudorandom $\mathrm{Exp}(1)$ noise for watermarking
- $w_{n+1} \leftarrow \arg\max_{y \in \mathcal{V}} \left\{ \dfrac{u_c(y)}{T} + E_c(y) \right\}$
- $n \leftarrow n + 1$

Algorithm 1 is distributionally equivalent to the original permute-and-flip procedure: randomly permute the vocabulary and return the first token whose Bernoulli flip succeeds with probability

$$\exp\left(\frac{u_c(y) - \max_{z \in \mathcal{V}} u_c(z)}{T} \right). \tag{2}$$

Without watermarking, one simply draws $E_c(y) \overset{\text{i.i.d.}}{\sim} \mathrm{Exp}(1)$. With watermarking, the same exponential variables are generated pseudorandomly from (k, c, y) as in (1).

2 Max-order PF: multi-draft speculative PF

Max-order PF uses B independent PF noise fields to generate B i.i.d. draft trajectories, while combining all B fields symmetrically into one exact target-PF noise field. For every context c, field $b \in [B]$, and token $y \in \mathcal{V}$, define

$$U_c^{(b)}(y) = \mathrm{PRF}_{\mathsf{k}}(c, b, y), \qquad E_c^{(b)}(y) = -\log U_c^{(b)}(y). \tag{3}$$

The b-th draft token is

$$X^{(b)}(c) = \arg\max_{y\in\mathcal{V}} \left\{ \frac{v_c(y)}{T} + E_c^{(b)}(y) \right\}. \tag{4}$$

For the target, let

$$R_c(y) = \min_{b\in[B]} U_c^{(b)}(y), \qquad \overline{U}_{c,B}(y) = 1 - \big(1 - R_c(y)\big)^B, \qquad \overline{E}_{c,B}(y) = -\log \overline{U}_{c,B}(y). \tag{5}$$

Since $\overline{U}_{c,B}(y) \sim \mathrm{Unif}(0,1)$ independently across y, the target token

$$Y_B(c) = \arg\max_{y\in\mathcal{V}} \left\{ \frac{u_c(y)}{T} + \overline{E}_{c,B}(y) \right\} \tag{6}$$

is exactly PF-distributed. Without watermarking, replace the keyed values in (3) by independent uniform random variables.

Some features of this method:

- **Exact PF marginals:** the B draft trajectories are i.i.d. draft-PF samples, while every emitted token follows the exact target-PF transition law.
- **Symmetric multi-draft coupling:** all B fields contribute symmetrically to the target noise; no draft field is chosen as an anchor.
- **Fixed-B drafter invariance:** the target rule uses all B fields and is independent of the draft logits and active draft set, so the drafter changes only the stopping time and computational efficiency.
- **Watermark preservation:** the detector regenerates $\overline{U}_{c,B}(w)$ from the key, so speculation does not alter the fixed-key Max-order PF watermark path or its detection score.
- **Simple GPU primitive:** we use uniform-race implementation for the exponential noises, and use fieldwise minima, multiplications, and argmin reductions, avoiding mapped Poisson cumulative clocks and global top-B selection comapreing to MPFR.

Algorithm 2: Watermarkable Max-order PF multi-draft speculative decoding

Given: lookahead K, number of homogeneous drafts B, output length N, target model P, draft model Q, initial prompt $w_{1:n}$, temperature T, and secret key k

while $n < N$ **do**
- $h \leftarrow w_{1:n}$; $L \leftarrow \min\{K, N-n\}$; $\mathcal{C}_0 \leftarrow \{h\}$; $\mathcal{S}(h) \leftarrow [B]$ `// active draft labels`
- `// Build` B `homogeneous PF draft trajectories and merge them into a tree`
- **for** $s = 0$ **to** $L-1$ **do**
  - $\mathcal{C}_{s+1} \leftarrow \varnothing$
  - **foreach** $c \in \mathcal{C}_s$ **do**
    - compute draft logits $v_c(\cdot) \leftarrow Q(c)$
    - **foreach** $b \in \mathcal{S}(c)$ **do**
      - regenerate $U_c^{(b)}(y) = \mathrm{PRF}_{\mathsf{k}}(c, b, y)$ for all $y \in \mathcal{V}$
      - $X^{(b)}(c) \leftarrow \arg\max_{y \in \mathcal{V}} \left\{ \frac{v_c(y)}{T} - \log U_c^{(b)}(y) \right\}$
    - $D(c) \leftarrow \{X^{(b)}(c) : b \in \mathcal{S}(c)\}$
    - **foreach** $y \in D(c)$ **do**
      - $\mathcal{S}(c \,\|\, y) \leftarrow \{b \in \mathcal{S}(c) : X^{(b)}(c) = y\}$
      - $\mathcal{C}_{s+1} \leftarrow \mathcal{C}_{s+1} \cup \{c \,\|\, y\}$
- `// Evaluate target PF tokens at all tree contexts in parallel`
- **foreach** $c \in \bigcup_{s=0}^{L} \mathcal{C}_s$ **do** `// the target always aggregates all` B `fields`
  - compute target logits $u_c(\cdot) \leftarrow P(c)$
  - **foreach** $y \in \mathcal{V}$ **do**
    - $R_c(y) \leftarrow \min_{b \in [B]} \mathrm{PRF}_{\mathsf{k}}(c, b, y)$
    - $\overline{U}_{c,B}(y) \leftarrow 1 - (1 - R_c(y))^B$
  - $Y_B(c) \leftarrow \arg\max_{y \in \mathcal{V}} \left\{ \frac{u_c(y)}{T} - \log \overline{U}_{c,B}(y) \right\}$
- $c^\star \leftarrow h$; *covered* $\leftarrow$ **true**
- **for** $s = 1$ **to** L **do**
  - $w_{n+1} \leftarrow Y_B(c^\star)$; $n \leftarrow n+1$ `// always emit the target Max-order PF token`
  - **if** $n = N$ **:**
    - **break**
  - **if** $Y_B(c^\star) \notin D(c^\star)$ **:**
    - *covered* $\leftarrow$ **false** and **break** `// the target path leaves the draft tree`
  - $c^\star \leftarrow c^\star \,\|\, Y_B(c^\star)$
- **if** *covered* ***and*** $n < N$ **:**
  - $w_{n+1} \leftarrow Y_B(c^\star)$; $n \leftarrow n+1$ `// precomputed target-PF bonus token`

3 Improved PF watermark detection

At each scored position t, let c_t be the context, let w_t be the emitted PF token, and define the recovered keyed pivot

$$V_t := \mathrm{PRF}_{\mathsf{k}}(c_t, w_t) = \exp\{-E_{c_t}(w_t)\}. \tag{7}$$

Under the null hypothesis that the observed text is independent of the secret key, $V_t \sim \mathrm{Unif}(0,1)$. For exact null calibration, we score only positions with distinct context labels, or otherwise use collision-aware calibration.

Original PF score. The detector of Zhao et al. [1] uses

$$T_{\mathrm{old}} = \sum_{t \in \mathcal{T}} -\log V_t. \tag{8}$$

If $M := |\mathcal{T}|$ pivots are independent under the null, then $T_{\mathrm{old}} \sim \mathrm{Gamma}(M, 1)$.

Li-style likelihood-ratio score. Let

$$\alpha_t(y) = \exp\left(\frac{u_{c_t}(y) - \max_z u_{c_t}(z)}{T}\right), \qquad \rho_t = \alpha_{t,(2)},$$

where $\alpha_{t,(2)}$ is the second-largest normalized PF weight. For a fixed reference value $\rho \in (0,1)$, define

$$f_\rho(v) = 1 - \rho v + \left(1 - \frac{v}{\rho}\right) \mathbf{1}\{v \le \rho\}, \qquad h_\rho(v) = \log f_\rho(v). \tag{9}$$

The statistic

$$T_\rho = \sum_{t \in \mathcal{T}} h_\rho(V_t)$$

is the exact log-likelihood-ratio statistic for the binary PF family with normalized weights $(1, \rho)$, and is the Li-style minimax score for the class in which the second-largest normalized weight is at least ρ. It is attractive when a reliable common lower bound ρ is available, but choosing one fixed ρ can be unnatural when reward or logit gaps vary substantially across prompts.

Lattimore-style adaptive power-law score. Motivated by the truncated power-law detector of Lattimore [3], we use the PF analogue

$$S_\epsilon(v) = \min\left\{v^{-1/2}, \epsilon^{-1/2}\right\} - (2 - \sqrt{\epsilon}), \qquad \epsilon \in (0,1). \tag{10}$$

The singularity is near 0, rather than near 1, because the PF watermark biases its selected pivot toward small values. Equivalently, for the selected exponential noise $Z = -\log V$,

$$S_\epsilon^E(Z) = \min\left\{e^{Z/2}, \epsilon^{-1/2}\right\} - (2 - \sqrt{\epsilon}).$$

The score is exactly centered under the null:

$$\mathbb{E}_0[S_\epsilon(U)] = 0, \qquad U \sim \mathrm{Unif}(0,1).$$

Given M scored positions, compute

$$T_\epsilon = \sum_{t \in \mathcal{T}} S_\epsilon(V_t),$$

and reject for large T_ϵ. The preferred level-δ critical value is the exact null quantile, estimated once by Monte Carlo from i.i.d. uniform pivots. A simple conservative analytic threshold is

$$\tau_\epsilon = \sqrt{2M \log\frac{1}{\epsilon}\log\frac{1}{\delta}} + \epsilon^{-1/2}\log\frac{1}{\delta}, \tag{11}$$

which satisfies

$$\mathbb{P}_0(T_\epsilon \geq \tau_\epsilon) \leq \delta.$$

For $0 < \epsilon \leq 1/16$, define the realized PF signal

$$G_\epsilon^{\mathrm{PF}} = \sum_{t\in\mathcal{T}} \min\left\{\sqrt{\rho_t}, \frac{\rho_t}{\sqrt{\epsilon}}\right\}. \tag{12}$$

The power-law detector succeeds with high probability once G_ϵ^{PF} is a sufficiently large constant multiple of τ_ϵ. In the homogeneous binary regime $\rho_t = \rho \geq \epsilon$, taking

$$\epsilon = \frac{\log(1/\delta)}{M} \qquad \text{when } M \geq 16\log(1/\delta)$$

gives the sufficient scaling

$$M\rho \gtrsim \log\left(\frac{M}{\log(1/\delta)}\right)\log\frac{1}{\delta}.$$

Any detector requires $M\rho = \Omega(\log(1/\delta))$ in this binary low-signal family. Thus the power-law detector is order-optimal up to logarithmic factors, while remaining model-agnostic and adaptive to heterogeneous ρ_t.

For comparison, the centered mean lift of the original score in the binary family is

$$\mathbb{E}_\rho[-\log V] - \mathbb{E}_0[-\log U] = \frac{\rho}{2}(1-\log\rho),$$

whereas the power-law score has mean lift of order $\sqrt{\rho}$ when $\epsilon \ll \rho$. We therefore use the power-law score as the default detector, the Li-style score as an oracle or structured benchmark when ρ is known, and the original PF score as a simple baseline.

References

[1] Xuandong Zhao, Lei Li, and Yu-Xiang Wang. Permute-and-Flip: An Optimally Stable and Watermarkable Decoder for LLMs. *International Conference on Learning Representations*, 2025.

[2] Xiang Li, Feng Ruan, Huiyuan Wang, Qi Long, and Weijie J. Su. A Statistical Framework of Watermarks for Large Language Models: Pivot, Detection Efficiency and Optimal Rules. *The Annals of Statistics*, 53(1):322–351, 2025.

[3] Tor Lattimore. Refined Detection for Gumbel Watermarking. arXiv:2603.30017, 2026.